Mathematics - Undergraduate Level Assessment 2 Total Marks: 30 Answer all questions.

## Multiple Choice (4 marks)

1. Statement 1 | If R is a ring and f(x) and g(x) are in R[x], then deg (f(x)+g(x)) = maximum (deg f(x)
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
2. Find the sum of the given polynomials in the given polynomial ring. f(x) = 4x - 5, g(x) = 2x^2 - 4x
   (a) 2x^2 + 5
   (b) 6x^2 + 4x + 6
   (c) 0
   (d) x^2 + 1

## True / False (16 marks)

*Answer True or False and justify briefly.*

1. True or False: The correct answer to 'Let C be the circle in the yz-plane whose equation is (y - 3)
2. True or False: The correct answer to 'Find the order of the factor group (Z_11 x Z_15)/(<1, 1>)'
3. True or False: The correct answer to 'Statement 1 | For any two groups G and G', there exists a h
4. True or False: The correct answer to 'Using Fermat's theorem, find the remainder of 3^47 when i
5. True or False: The correct answer to 'Determine whether the polynomial in Z[x] satisfies an Eisen
6. True or False: The correct answer to 'What is the area of an equilateral triangle whose inscribed
7. True or False: The correct answer to 'A tree is a connected graph with no cycles. How many non
8. True or False: The correct answer to 'Statement 1 | If H is a subgroup of G and a belongs to G th

## Long Form Response (10 marks)

*Provide thorough reasoning and reference key steps.*

1. Expand the following expression: $3(8x^2-2x+1)$.

*End of Paper*